

PUDASJARVI AP, Finland

WMO: 028660

Lat: 65.414N

Lon: 26.963E

Elev: 120

StdP: 99.89

Time Zone: 2.00 (EUE)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.1	(c) -27.0	(d) -33.1	(e) 0.2	(f) -30.0	(g) -29.7	(h) 0.2	(i) -26.9	(j) 8.4	(k) -4.4	(l) 7.4	(m) -4.8	(n) 0.8	(o) 80	(p) 0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	26.0	16.9	23.9	16.1	22.0	15.3	18.8	22.9	17.6	21.9	16.5	20.5	3.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.5	20.8	15.9	11.4	19.3	14.7	10.6	18.2	53.8	22.9	49.9	22.2	46.5	20.6	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.6	6.6	5.9	-33.8	27.8	3.3	2.6	-36.2	29.6	-38.1	31.2	-40.0	32.7	-42.4	34.6
			-33.9	20.0	3.3	1.7	-36.3	21.2	-38.2	22.2	-40.0	23.2	-42.4	24.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-10.9	-10.4	-6.2	0.7	7.7	12.6	16.2	13.8	8.6	1.9	-3.2	-7.3		
	DBStd	10.72	7.91	7.66	5.89	4.08	4.57	3.69	3.39	3.35	3.66	4.31	5.59	7.47		
	HDD10.0	3378	648	570	502	280	101	16	1	7	69	251	396	538		
	HDD18.3	5977	907	803	760	529	332	176	83	144	293	508	646	796		
	CDD10.0	469	0	0	0	1	28	93	194	126	26	1	0	0		
	CDD18.3	27	0	0	0	0	1	4	17	5	0	0	0	0		
	CDH23.3	234	0	0	0	0	16	40	146	33	0	0	0	0		
	CDH26.7	36	0	0	0	0	1	7	24	4	0	0	0	0		
(14) Wind	WSAvg	2.8	2.3	2.6	2.9	2.9	3.2	3.1	2.7	2.7	2.8	2.8	2.7	2.6		
(15) Precipitation	PrecAvg	617	46	40	33	30	54	58	82	71	59	54	52	50		
	PrecMax	756	85	82	52	65	107	106	136	113	107	98	81	85		
	PrecMin	524	7	11	13	8	20	21	11	19	23	8	14	12		
	PrecStd	74	26	19	12	18	25	26	33	23	23	22	19	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	4.2	6.6	15.0	25.4	27.5	29.3	26.8	20.0	12.1	7.7	4.0		
		MCWB	1.9	2.6	3.0	8.6	13.3	16.9	18.7	18.9	14.8	10.3	7.0	3.5		
	2%	DB	1.1	2.1	4.2	11.2	22.2	23.8	26.9	23.7	17.3	10.0	6.2	2.8		
		MCWB	0.7	1.3	1.6	5.9	13.0	15.6	17.7	16.7	13.3	8.7	5.6	2.3		
	5%	DB	0.2	0.6	2.7	8.9	18.5	21.5	24.9	21.8	15.7	8.7	4.4	1.7		
		MCWB	0.0	0.2	0.9	4.6	11.2	13.7	16.8	16.0	12.4	7.7	3.9	1.3		
	10%	DB	-0.7	-0.5	1.4	6.8	15.7	19.7	22.8	19.9	14.2	7.5	3.0	0.7		
		MCWB	-1.0	-0.9	0.1	3.2	9.8	12.9	16.1	15.1	11.9	6.7	2.7	0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.7	3.4	9.5	15.3	18.5	21.0	19.9	15.6	10.7	7.3	3.7		
		MCDB	2.6	3.8	5.6	13.5	21.6	23.4	26.0	24.4	18.4	11.5	7.6	3.9		
	2%	WB	0.8	1.4	2.1	6.6	13.7	16.8	19.4	18.1	14.1	9.1	5.6	2.3		
		MCDB	1.0	2.0	3.6	10.5	20.3	22.6	23.8	22.3	16.6	9.7	6.1	2.6		
	5%	WB	0.1	0.2	1.2	4.8	11.9	15.2	18.3	16.8	13.1	7.8	4.0	1.3		
		MCDB	0.3	0.6	2.3	8.2	18.2	19.6	22.5	20.1	15.0	8.5	4.4	1.6		
	10%	WB	-1.0	-0.9	0.4	3.6	10.3	13.8	17.1	15.7	12.1	6.7	2.7	0.5		
		MCDB	-0.7	-0.6	1.3	6.2	14.7	18.0	21.3	18.9	13.9	7.5	3.0	0.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.7	9.6	8.5	9.3	9.0	9.2	8.8	7.3	4.8	4.6	6.1		
		MCDBR	5.3	5.3	8.0	11.5	14.7	12.6	12.3	12.0	9.5	5.3	4.3	4.2		
	5% WB	MCWBR	5.2	4.8	6.3	7.2	7.1	5.5	4.9	6.1	5.9	4.4	4.2	4.0		
		MCWBR	5.2	5.1	6.3	10.3	13.1	10.4	10.0	9.7	8.3	5.2	4.2	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.201	0.271	0.302	0.338	0.343	0.344	0.363	0.344	0.321	0.296	0.347	N/A		
	taud		1.937	2.108	2.164	2.253	2.453	2.472	2.435	2.471	2.524	2.368	2.620	N/A		
	Ebn at Noon		277	595	759	825	865	870	837	820	763	586	267	N/A		
	Edh at Noon		21	58	88	104	94	95	96	84	64	46	20	N/A		
(44) All-Sky Solar Radiation	RadAvg		0.10	0.60	2.02	3.72	4.71	5.20	4.89	3.59	1.95	0.69	0.15	0.02		
	RadStd		0.01	0.11	0.20	0.34	0.46	0.41	0.49	0.35	0.18	0.11	0.03	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page